Climate Policy

ISSN: 1469-3062 (Print) 1752-7457 (Online) Journal homepage: www.tandfonline.com/journals/tcpo20

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To cite this article: Yann Briand, Johannes Svensson, Clara Lepin, Henri Waisman, Christopher Bataille, Lukytawati Anggraeni, Vidhee Avashia, Aliosha Nicolás Behnisch, Rizaldi Boer, Daniel Buira, Savahna De Kock, Retno Gumilang Dewi, Carolina Burle Dubeux, Amit Garg, Thalia Hernandez, Francisco Lallana, Emilio Lèbre La Rovere, Bryce McCall, Marina Recalde, Ucok W.R. Siagian, Saritha Sudharmma Vishwanathan, Julia Tatham, Fei Teng, Jordi Tovilla & William Wills (2026) Net-zero compatible development pathways in Argentina, Brazil, China, India, Indonesia, Mexico and South Africa: lessons for short-term actions, Climate Policy, 26:5, 709-725, DOI: 10.1080/14693062.2026.2624246

To link to this article: https://doi.org/10.1080/14693062.2026.2624246

Published online: 10 Feb 2026.

CLIMATE POLICY
2026, VOL. 26, NO. 5, 709–725
https://doi.org/10.1080/14693062.2026.2624246

Taylor & Francis
Taylor & Francis Group

RESEARCH ARTICLE

Net-zero compatible development pathways in Argentina, Brazil, China, India, Indonesia, Mexico and South Africa: lessons for short-term actions

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ABSTRACT

The global carbon neutrality objective is a key upshot of the Paris Agreement climate goals. Its translation to the national level has been central to climate policy since then and, notably, country commitments have flourished over the years. However, there is still limited scientifically robust and policy-relevant evidence on national pathways to net-zero that captures the specifics of national circumstances. Such insights would be critical to ensure that carbon neutrality pledges are credible and actionable. This paper presents the methodological approach developed under the Deep Decarbonization Pathways (DDP) initiative to support the elaboration by in-country experts of national scenarios exploring long-term transformations consistent with carbon neutrality, and to enable a constructive dialogue on carbon neutrality with a large set of decisionmakers. We present carbon neutrality scenario results for 7 large emerging economies developed with this method. They display how each country can define its own pathway to net-zero, considering its national circumstances and in a way that preserves key domestic socio-economic priorities. They display some country-specific patterns but also highlight key common transformations for carbon neutrality, including moderation of final energy uses, a steady decrease in the use of fossil fuels for all purposes and substitution of low GHG energy forms, and a steady increase in the absorption capacities in diverse land uses. To trigger these net-zero transformations, national public decision makers should evolve national development policy packages according to the three following policy insights.

Key Policy Insights

- Update existing economic and non-economic policy instruments in sectors where commercially available and low-carbon solutions exist and most short-term emission reductions are available: power generation, passenger transport, and the land use sector.

ARTICLE HISTORY
Received 28 March 2025
Accepted 26 January 2026

KEYWORDS
carbon neutrality; country-driven strategies; sectoral transformations; energy transition; land use sinks; policy packages

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Supplemental data for this article can be accessed online at https://doi.org/10.1080/14693062.2026.2624246.

- Analyse inertias in national net-zero transformations and implement measures to avoid lock-ins, develop and encourage new infrastructure, technologies, lifestyles and institutions that are necessary to achieve deeper and longer-term emission reductions in all sectors. It reinforces the importance of having a long-term policy perspective when taking short-term decisions.
- Adopt measures in climate policy packages to manage the socio-economic challenges of the transition and accompany structural changes in the economic systems to ensure a desirable transition to net-zero.

1. Introduction

The Paris Agreement (Art 2.1a) established a global climate goal of 'holding the increase in the global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5°C above pre-industrial levels'. Scientific assessments conclude that global CO_2 emissions should reach net-zero between the early 2050s and the early 2070s and then become negative (statement C2 in *IPCC, 2022*). This does not require, however, that all countries reach net-zero by the same date and the same way. However, this goal would become unattainable if any major emitter was off track and if certain jurisdictions do not go net negative. An accurate translation of this global goal to the national scale, while mindful of national circumstances, is therefore critical.

This translation from global to national has been central to both international and national climate policy since the Paris Agreement was reached. The carbon neutrality objective has proved its relevance as a political guidepost, anchoring domestic ambition against scientifically grounded benchmarks. Today, 108 countries, representing 82% of global emissions, have communicated a net-zero pledge. However, there is a persistent gap between observed emission trajectories and extrapolations based on current policies and the emission levels needed to reach net-zero & negative in the long-term (UNEP, 2024).

This yawning deficit in net-zero implementation highlights a crucial challenge that countries face when designing domestic actions and targets. The net-zero goal requires unprecedented and profound systems' change as compared to past and current trends, i.e. structural transformations in the socio-economic organization of energy, industry, urban, building, transport, agriculture and land use systems (Statement C4-C9 in IPCC, 2022). To ensure that the net-zero goal is credible and actionable and that progress towards this goal can be monitored, these systemic changes towards net-zero must be turned into concrete decisions with clear milestones over time towards the net-zero date. To add to the complexity, relationships between the systems must be recognized, as well as the socio-economic implications of transformative change. Economy-wide packages tailored to national circumstances are therefore necessary (see statement E4 in IPCC, 2022; Dubash et al., 2022).

Long-Term Low Emission Development Strategies (LT-LEDS), introduced in article 4.19 of the Paris Agreement (UNFCCC, 2015) and reiterated in articles 32–35 of the 2021 Glasgow Climate Pact (UNFCCC, 2021), and in Arts 40 & 42 of the Global Stocktake decision at COP28 (UNFCCC, 2023), are key instruments to support alignment between short-term actions and long-term goals (Català et al., 2024). LT-LEDS could help countries and actors explore pathways towards carbon neutrality, define a clear sequence of actions towards the long-term objective, and help identify robust short-term decisions in a context of deep uncertainty. However, current research on long-term strategies is dominated by global integrated assessment models (IAMs), which, beyond their strengths to capture the temperature consequences of global emission pathways, have several limitations in capturing nationally driven political, social, and institutional feasibility conditions (de Bortoli et al., 2025; Muttitt et al., 2023).

Consequently, since 2013 the Deep Decarbonization Pathways (DDP) network has developed methods for creating country-driven pathways designed to exploit the full potential of LT-LEDS for supporting national actions consistent with the Paris Agreement climate and domestic development goals (Bataille et al., 2016, 2020; DDPP, 2015). One key feature of network members stands on their capacity, as in-country policy researchers, to carefully include their national political, economic, technological, socio-cultural and geographic context, and to use the same methodology to design climate policy pathways and compare them. This article aims to discuss the commonalities and specificities of seven large emerging economies to reach carbon neutrality by

2050–2070 and their socio-economic priorities, as well as to extract some cross-country and short-term policy lessons. The paper includes analysis from Argentina, Brazil, China, India, Indonesia, Mexico, South Africa, building on the most recent research of the DDP network through the IMAGINE project.

The remainder of the paper is structured as follows. Section 2 presents the DDP country analysis method and discusses how carbon neutrality led to revision of the approach compared to earlier versions. Section 3 presents the key characteristics of the national pathways to net-zero, while Section 4 discusses short-term policy implications of these analyses. Section 5 concludes. Sector-specific deep dives of these methods, results and policy dimensions can be found in other papers of this Special Issue.

2. The deep decarbonization pathways method

2.1. Introduction

This research uses the DDP approach and method for pathway design. It is grounded on a set of key principles presented in *Bataille et al. (2016) and Waisman et al. (2019),* which articulates national narratives, models and reporting in an iterative scenario design process. Key principles include: backcasting from the goal and sectoral specific 'attractors' to explore the long-term in a way to inform short-term decisions (Robinson, 1982); the work is led by in-country research teams, who are free to choose their assumptions and models thanks to a model-agnostic framework (Pye & Bataille, 2016); the method is suggestive but not prescriptive of a wide range of mitigation options, drivers, and co-benefits beyond those that may be represented in domestic models; and finally the method supports a structured, ideally iterative engagement with real-world decision makers and other stakeholders using non-technical narratives and providing key quantitative indicators readily identifiable to stakeholders.

This pathway design framework has been used in past DDP projects and is also used as the framework for this study. However, the methodological challenges raised by carbon neutrality lead to a set of refinements in each of the components of the framework.

2.2. Methodological challenges related to carbon neutrality

To explore carbon neutrality, it is necessary to consider at least three main methodological challenges. First, carbon neutrality means a 'balance between anthropogenic emissions by sources and removals by sinks' (Art 4.1. in *UNFCCC, 2015*), requiring detailed analysis of carbon flows to and from the Agriculture, Forestry and Land Use (AFOLU) sector. This requires detailed accounting of all related drivers, such as land use competition, changes in agricultural practices, trade-offs with biodiversity or socio-economic impacts (Nabuurs 2022; Svensson et al., 2021).

Second, it requires consideration of all possible mitigation options, especially those affecting carbon fluxes (statement C3 in IPCC, 2022), going well beyond the only consideration of zero-GHG emission technologies. It requires consideration of long-term changes in organizational patterns and demand shifts across the economy (Statement C10 in IPCC, 2022; Barrett et al., 2022; Creutzig et al., 2018, 2021; Fisch-Romito et al., 2021; Wiese et al., 2024).

Finally, net-zero and negative GHG flows require overcoming the scepticism about the feasibility and desirability of very profound transitions in a given country context, especially in a developing country context. The need for well-designed, communicated, and fully discussed cross-sectoral policy packages on different time horizons adds to the challenge because it requires scenario techniques that capture the main dynamics but in a way that still enables a non-technical dialogue with a large set of decisionmakers (Braunreiter et al., 2021).

2.3. Major refinements of the pathways design framework for carbon neutrality

The DDP pathway design and reporting framework has been extended beyond the analysis of the energy system and mostly CO_2 emissions to support a broader analysis of all emission sources and sinks. This means concretely that the framework covers all sectoral sources and sinks, with explicit representation of six energy consumption sectors,[1] three energy production sectors,[2] and a dedicated representation of AFOLU and Waste sectors. Some of these sectoral decompositions were already used in the DDP-LAC project but were refined for this study (Lefèvre et al., 2020; Svensson et al., 2021). Others were absent from previous projects and introduced for this study given the new focus on these sectors in this project (Bataille et al., 2023; Briand et al., 2024). In addition, two new thematic tabs have been added in the framework to represent the international policy context on the one hand and provide more details on the socio-economic context on the other hand.

Concretely, each sectoral and economy-wide component is described in the pathways design framework with a dedicated storyline and a dedicated dashboard. The following sub-sections provide a synthesis of the main methodological enhancements that were implemented for this study in each sub-component of the framework.

2.3.1 Storyline

The ***storyline*** organizes the scenario design around mixed qualitative and quantitative descriptions of the key drivers of decarbonization, providing a clear underpinning for the assumptions used in the modelling (Annex 1). It also integrates dimensions of the scenarios that cannot be modelled explicitly but form an integral part of the overall scenario.

Each storyline is now organized according to a systematic structure in chapters that distinguish the main categories of drivers. An effort has been made to ensure a comprehensive accounting of all key sectoral drivers of the decarbonization transition, exploring not only technical aspects but also, organizational and structural dimensions as relevant.

The content of each chapter is guided by concrete questions to ensure comparability of information reported in different countries. Each chapter is designed to capture different actors' views, acknowledging that various actors have to drive changes across different and segmented sectoral components. As each country has a different socio-organizational policy context, the storyline framework is a key flexible instrument for organizing the policy engagement with a diverse set of stakeholders around the feasibility and desirability of scenarios.

All storylines are also structured around two key time periods, between now and 2030–2035 and between 2030–2035 and 2050–70. This facilitates the articulation of measures and changes to implement by the horizon of the nationally determined contributions (NDCs), and those to implement after 2035 towards the national carbon neutrality date.

2.3.2 Dashboard

The ***dashboard*** organizes a quantitative representation of the scenario based on key indicators, chosen because the quantification can be done in a robust manner and because it can provide a more tangible vision of the transition for decisionmakers (Annex 1).

Selected for their policy relevance, several indicators have been added to capture some stakeholder-oriented drivers and policy priorities critical to the carbon neutrality transition. This concerns indicators which measure structural, profound changes in the economy, and the new models of development and organization that are required to reach carbon neutrality.

All dashboards have a detailed representation of the CO_2 and other relevant GHG sources and sinks, as well as the energy demand and production. Beyond emissions and energy accounts, an enhanced granularity of indicators characterizes impacts on socio-economic development, changes in physical assets and infrastructures, as well as changes in demand for mobility, industrial materials and buildings, among others.

New indicators and systematic tests have also been introduced in the reporting structure to allow systematic analysis of the interplay between different segments of the scenarios and ensure internal consistency between economy-wide and sectoral tabs. As such, the dashboards also provide a framework for checking the consistency between the national and the sectoral transformations.

[1]Passenger transport, Freight transport, Energy & Process Emissions Intensive Industries, Light Industries, Residential buildings, Commercial buildings and public services.

[2]Extractive energy industries, Power production, Other energy industries.

2.3.3 Analytical methods, including modelling

The DDP pathway design and reporting framework is open to operating with any type of model or quantification instrument to inform dashboards, with any type of social science methodologies to develop storylines and can also be informed by expert judgement. Models help translate the storylines into quantified information in the dashboard, but the models do not limit the definition of the storylines. They also strengthen the consistency between different dimensions of the narratives by capturing complex interrelationships. As such, they can be instrumental in the iterative process of pathways design, notably by revealing indirect effects that may not have been anticipated and therefore suggesting revisions of narrative assumptions.

In-country research teams have focused their model developments on the strengthening of their capacity to assess specific components of the narratives and systemic relationships of crucial importance for their national policy process and stakeholder engagement (Annex 2). Different country teams may use different analytical methods to inform a given dashboard indicator. Transparency on this information is key to ensuring proper interpretation of the dashboard and to accurately understand the differences that may arise from assessing a given parameter with different analytical methods in cross-country comparisons.

3. Results – National deep decarbonization pathways

This part presents modelled results of pathways over 2020[3]–2050/70 as designed with the above-described methodology for seven emerging economies: Argentina, Brazil, China, India, Indonesia, Mexico and South Africa.

3.1. Definition of scenarios

Two main families of scenarios are considered in each country to assess the gap between current public and private actions and what is required to align with the Paris Agreement goals. The Current Policy Scenarios (CPS) reflect the pursuit of current trends and the integration of existing or planned policies and actions within a country. The Deep Decarbonization Scenarios (DDS) explore ambitious emission reductions aligned with the Paris Agreement goals and consistent with key socio-economic priorities. The analysis will focus on the DDS, using CPS as a reference point for national ambitions.

In most countries, the DDS are run under the assumption of a growth in population (except in China) and a growth of GDP per capita. However, as GDP per capita is largely insufficient to characterize the socio-economic needs, each country team has selected context-specific priorities (Table 1). This selection was done by each local research team given their own experience and understanding of the key societal concerns of their country (Annex 3).

The definition of the DDS is based on official political targets when countries have one: climate neutrality by 2050 in Brazil (Government of Brazil, 2024); CO_2 neutrality by 2050 in South Africa (Government of South Africa, 2020); net-zero by 2060 in China (Government of China, 2021); net-zero emissions by 2060 in Indonesia (Government of Indonesia, 2021); net-zero emissions by 2070 in India (MoEFCC, 2022a). The formulation of the target is sometimes equivocal regarding the gases considered, which has led to interpretations at the country level by in-country partners.

In Mexico, even in the absence of an official net-zero commitment, the DDS also considers net negative CO_2 emissions by 2050, with a carbon neutrality date around 2045. Finally in Argentina, the scenario is modelled only until 2050, and a simple extrapolation leads to carbon neutrality by 2070; in the absence of an official governmental position, this trajectory is an expert-based judgement of the national perspective regarding the potential for decarbonization in all sectors, including key hard-to-abate sectors (notably industry and freight sectors). Annex 4 provides more detailed information on these scenarios, including the overall narrative of the CPS and DDS scenarios considered, and publicly available references for further details on the country analyses.

In the rest of the analysis, we focus on CO2 emissions, as the main greenhouse gas conditioning climate stabilization in the long-term. Other greenhouse gases are also important for reaching the goals of the Paris

[3]The calibration dates for these models vary by country, ranging from 2015 to 2022, depending on the availability of national datasets. To ensure comparability, extrapolation ensures a common starting date in 2020 for all figures and quantitative analyses. For Brazil, the 2020 values are mostly based on actual data rather than model estimates.

Table 1. Repartition per country of socioeconomic effects studied (Y – Yes in green or N – No in red).

	Argentina	Brazil	China	India	Indonesia	Mexico	South Africa
Economic Situation & Inequalities	Y	Y	Y	Y	Y	Y	Y
Demographic Situation	Y	Y	Y	Y	Y	Y	Y
Job Market & Human Capital	Y	Y	N	N	Y	Y	N
Energy Access	N	N	N	Y	Y	N	Y
Food Security	N	N	N	Y	Y	N	N
Quality of Life in Urban Areas	N	N	N	N	N	Y	Y
Air Quality & Pollution	N	N	N	N	Y	N	Y
Land use: biodiversity and indigenous communities	N	Y	N	N	N	N	N

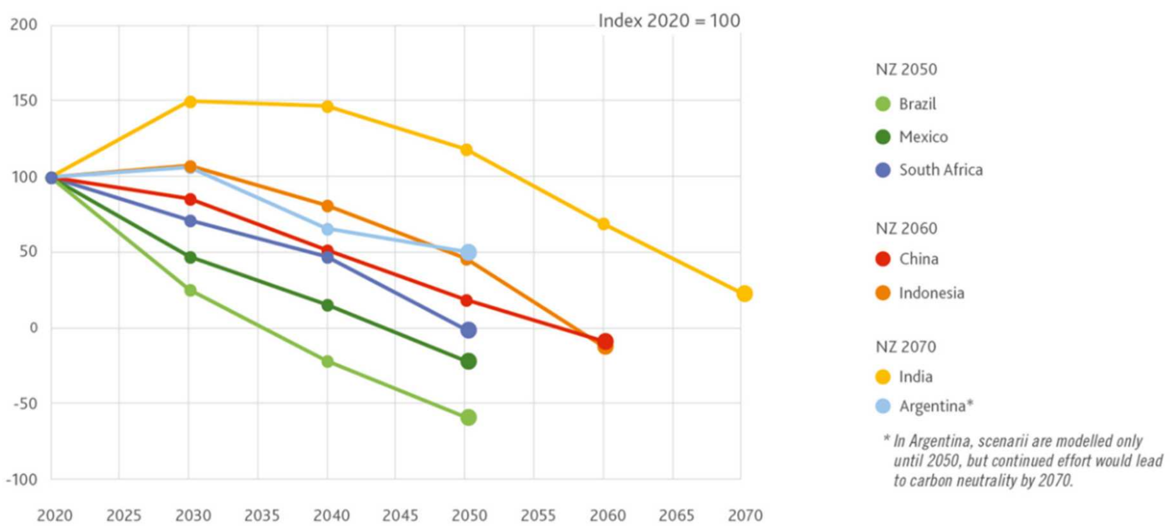

Figure 1. National net-CO_2 emissions, 2020-50, Index.

Agreement, but only a few country analyses have conducted an in-depth assessment of these gases, so we do not discuss them in this cross-country paper.

3.2. Economy-wide results

The results show that the DDS reaches zero, and then net negative, carbon emissions over 2020–2070 (Figure 1**)**. Net negative emissions often correspond to countries where the net-zero target concerns GHG emissions so that net negative CO2 emissions are needed to compensate for remaining non-CO2 gases. In Argentina, the CO2 emissions are on a downward track, but the analysis is not extended beyond 2050 when carbon neutrality could be reached. In India, the analysis has been extrapolated beyond 2050 based on *Garg et al. (2024)* and net-zero carbon emissions are not reached by 2070 but would be reached by around 2075.

The trajectory of total CO2 emissions displays very different patterns across countries**,** which can be partially explained by the different starting points and target dates for net-zero but also result from differences in the country's strategies. To explore these drivers, we distinguish the CO2 emissions trends from combined energy and industrial process and product use (IPPU) on the one hand, and from the land use, land use changes and forestry (LULUCF) on the other hand, as the two dominant sources of CO2 emissions (Figure 2).[4]

[4]We ignore notably agriculture and waste from this analysis, given the very minor role in CO2 emissions across the countries considered in this analysis.

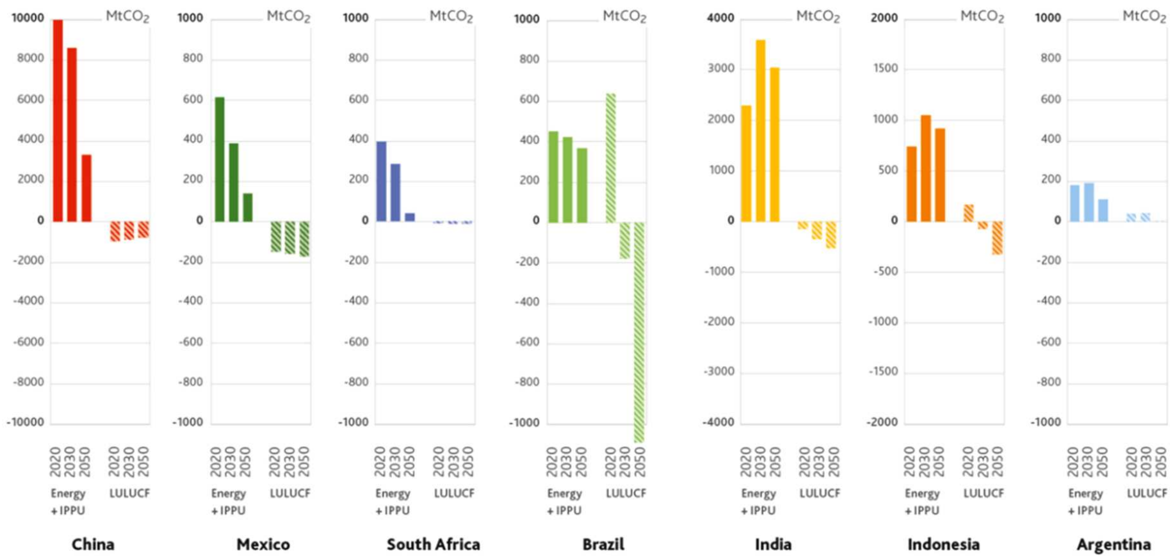

Figure 2. CO_2 emission sources and sinks by countries and sectors, 2020-50, $MtCO_2$.

In several countries (Brazil, China, Mexico, South Africa), emission trends display an immediate and continuous decrease of total CO2 emissions. In these cases, the main driving force is falling energy-related CO2 emissions, which is the dominant source of CO2 emissions and display a strong decrease throughout the period. A noteworthy counterexample can however be found in Brazil, where the major source of CO2 emissions in 2020 is from the LULUCF sector. In Brazil the steep decrease is triggered by massive land use sector efforts, becoming a deep carbon sink by 2050, while energy-related emissions decrease only moderately (about – 20%).

Some country pathways (Indonesia and Argentina) display a slight increase over the first decade before a steady decline. This trend is even more marked in India. These correspond to situations where energy consumption increases to satisfy industrial and demographic needs at a faster rate than energy efficiency and decarbonization options are implemented in sectors. In all cases, a reversal of this trend happens in the medium term combined with an acceleration of the emission reductions in the LULUCF sector.

The shape of the curve is largely driven by the carbon neutrality date, however, country specificities still play a role. For example, China and Indonesia have the same carbon neutrality date but slightly different profiles, given notably different stages of development and technological progress (Indonesia is less advanced and therefore requires an increase of energy-related emissions) and different roles for the LULUCF sector (in Indonesia, the sector turns from a net emitter to an important net sink, while in China it stays at a rather stable negative level throughout the period).

3.3. Energy system changes

The deep emission reductions from energy uses and industrial processes are the result of changes to moderate energy demand per capita while improving the quality of life (see Section 3.3.1.) and of changes to decrease the use of fossil fuels and replace it with renewable energies (see Section 3.3.2.).

3.3.1 A moderation of final energy uses

Even though energy consumption per capita increases in absolute terms from 2020 to 2050 in most countries in the DDS, by 2050 it remains below the levels of industrialized countries today (cf. the US level at around 200 GJ/capita in 2020). In addition, we can note that energy consumption is lower in DDS compared to CPS in all countries except India by 2050 (Figure 3). In the case of India and Indonesia, the very low starting point in 2020 explains mostly why the difference between CPS and DDS is limited. These two countries have done a quick catch-up

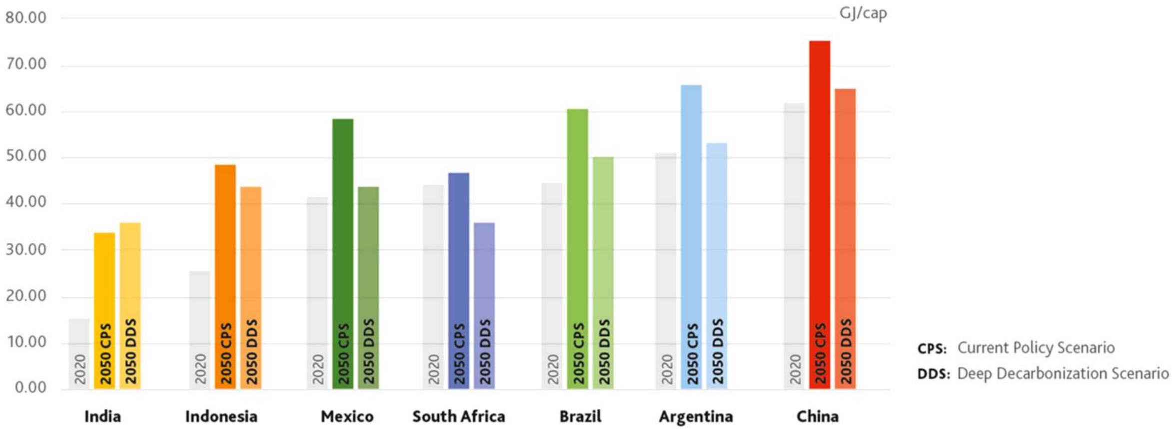

Figure 3. Final energy consumption per capita, CPS vs DDS, 2020-50, GJ/cap.

of energy demand levels by 2050 in parallel with a deep decarbonization trajectory. Comparatively, South Africa is the only country with an absolute decrease of energy consumption per capita between 2020 and 2050 in DDS. This is due to the saturation of energy access and the dissemination of energy-efficient technologies.

While all countries simulate enhanced access to services which drives up energy uses, other transformations lead to a moderation of energy demand by 2050 between CPS and DDS. This is mostly due to the deployment of energy-efficient systems and less energy-intensive organizations, practices and lifestyles. In all countries, the transport sector plays a critical role in the moderation of demand between DDS and CPS by 2050, which highlights that demand control strategies are critical to the deep decarbonization of this sector. This is particularly the case for passenger transport in Brazil, China, Mexico and freight transport in South Africa (Briand et al., 2023, 2024). In other countries, the reduction of energy consumption in the industrial sector like in Indonesia (Bataille et al., 2023), or in the residential buildings like in Argentina and India, are key. This shows that net-zero pathways, integrating country-specific development objectives, can help identify development patterns with a lower energy input.

3.3.2 A continuous decrease in the direct use of fossil fuels

The reduction of the carbon content of energy is enabled by a continuous reduction of fossil fuel use in final energy consumption (Figure 4(a)), being replaced by electricity and biomass energy mostly, and a rapid decrease of fossil fuel use in electricity generation (Figure 4(b)).

In most countries, fossil fuels account for approximately 70% of the final energy consumption and between 55% and 80% of the electricity generation in 2020, except in Brazil which already has a very important use of biomass energy for solid and liquid fuels and hydroelectricity. DDS highlight the need for a significant decline in the use of fossil fuel in final energy consumption and in power generation in all countries. By 2050 in most countries, fossil fuels will fall in a range of around 20–30% of final energy consumption and around 0–5% of power generation, except in India and Indonesia (Figure 4).

The shape varies by country according to the specificities of the socio-economic context and energy landscape, the target dates for national carbon neutrality and the pressure on the energy sector given negative emission potential in the land use sector. For example, in Indonesia and India, similar levels of fossil fuels in final energy are reached, but respectively by the date of their carbon neutrality, 2060 and 2070. For other countries after 2050, fossil fuel use will continue to decrease to reach for example less than 10% of final energy, like in China. In comparison, Brazil and Indonesia experience a more moderate decrease, notably because their LULUCF negative emission potential can compensate for higher residual emissions from fossil fuel use. For power generation, the rapid decrease and replacement of fossil fuels by renewables is a common feature for all countries, which will be almost all fossil-free by their dates for national carbon neutrality.

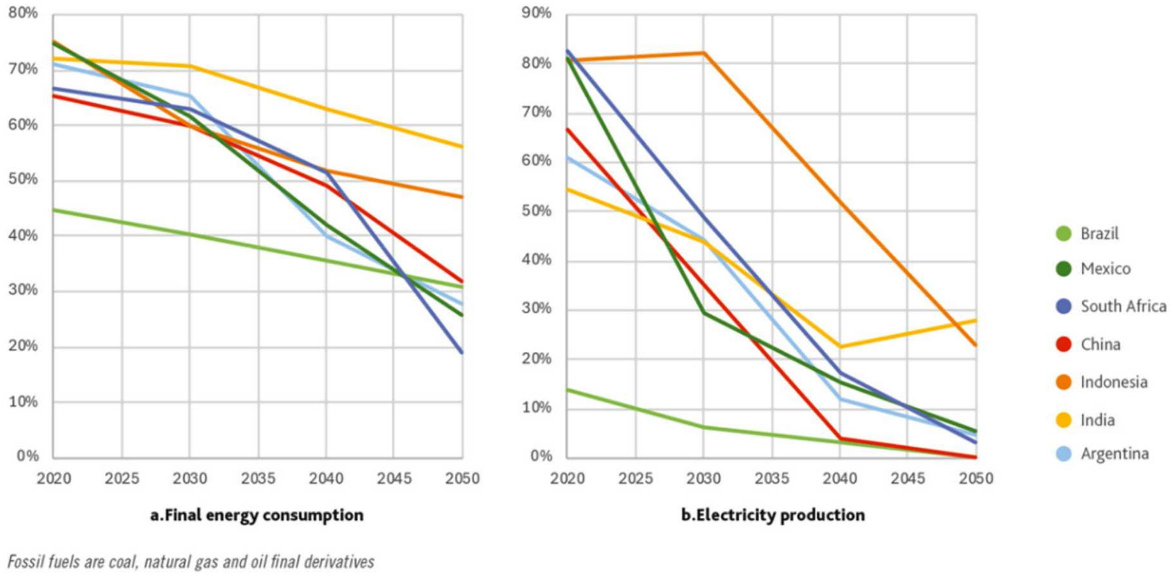

Figure 4. Share of fossil fuels in final energy consumption (a-left) and in electricity production (b-right).

In India, we can note a slight increase over the period 2040–2050 due mainly to some natural gas stranded assets developed before 2040 for the rapid expansion of electricity access and grid stability.

In absolute terms, most countries show a reduction of absolute use of fossil fuels in final energy consumption, except India and Indonesia, where fossil fuel use eventually declines after 2050 (Annex 4).

3.4. Land use system changes

Deep emission reductions from the land use systems (LULUCF) come mostly from distinct changes to increase CO_2 absorption of forests (Section 3.4.1) and of crop – and grasslands (Section 3.4.2). In some cases, measures to reduce the degradation of peatland could also be significant as in Indonesia (Annex 4, Svensson et al., 2024). Argentina and China have not developed a detailed analysis of the land use sector in their scenarios and are therefore not included in this section.

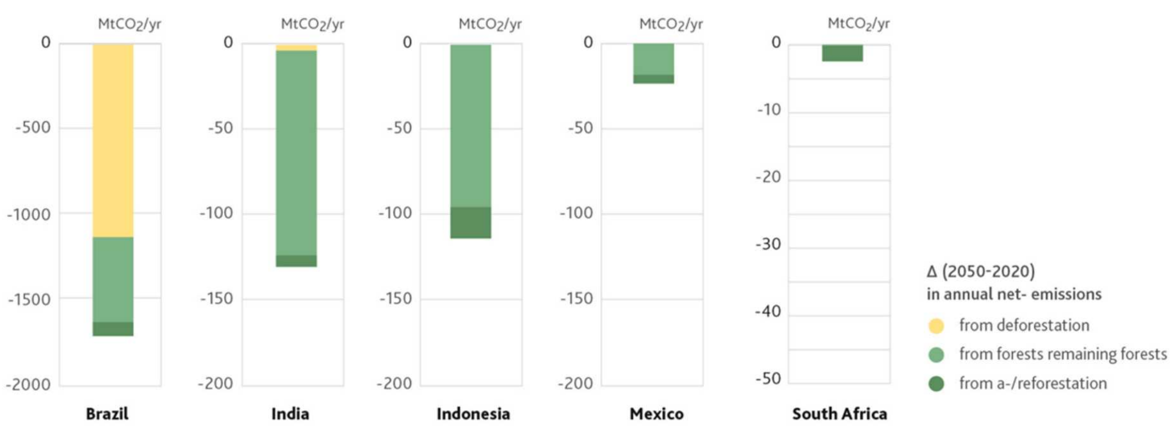

Figure 5. Difference between annual net-emission reductions from forests in 2050 and 2020, $MtCO_2$.

3.4.1 An increase in absorption capacities in forestlands

The increase in CO2 absorptions in forestlands is driven by the increase in the forest surface through reductions in the rate of deforestation and increases in the rates of a-/reforestation, and by improved forest management of existing forests (Figure 5).

A reduction in the rate of deforestation is the most important forest-related driver in Brazil, whereas a-/reforestation is the most important driver in India, Indonesia and Mexico. Increasing annual sequestrations from remaining forests plays a smaller role in most countries.

Following reductions in deforestation rates and increases in a-/reforestation, the share of the national territory under forest cover increases between 2020 and 2050 in Brazil (from 60 to 61%), India (from 22% to 25%) and Indonesia (from 50% to 56%). A key driver of deforestation in Brazil and Indonesia is agricultural expansion. To reduce deforestation, and to make more land available for a-/reforestation, both countries transform agricultural production to increase agricultural productivity (see Section 3.4.2).

3.4.2 An increase in absorption capacities in crop-and grasslands

The increase in CO2 absorption in crop – and grasslands is mostly driven by increased sequestration in croplands, especially in India, while grasslands play a smaller role in most countries and even remains a net emitter in Brazil and Indonesia (Figure 6). This is driven by an increase in the sequestration capacity per hectare of croplands and by an increase in total cropland surfaces like in Brazil and Indonesia.

Changes in agricultural practices are the main drivers to increase sequestration and storage of CO2 of croplands. It includes agroforestry and practices that enhance soil carbon, such as crop rotation and reduced tillage practices, the application of organic manures and compost, and the application of biochar. For grasslands, there is no or little shift in practices to increase the sequestration of carbon. Some of these practices can contribute to increasing agricultural productivity without synthetic fertilizers to meet a growing demand while freeing up lands for forest expansion (Annex 4, Svensson et al., 2024).

3.5. Residual emissions in 2050

In 2050, all countries studied have net-zero or net negative LULUCF emissions, but residual CO2 emissions coming from the use of fossil fuels in the sub-sectors of the energy system and from industrial processes (IPPU) (Figure 7). Most remaining emissions in 2050 come from the industrial and the transport sector, discussed in Bataille et al. (2023) and Briand et al. (2023, 2024), but depending on country context, other main remaining sources could come from buildings, like in China, or power, like in India, Indonesia or Mexico (Figure 7).

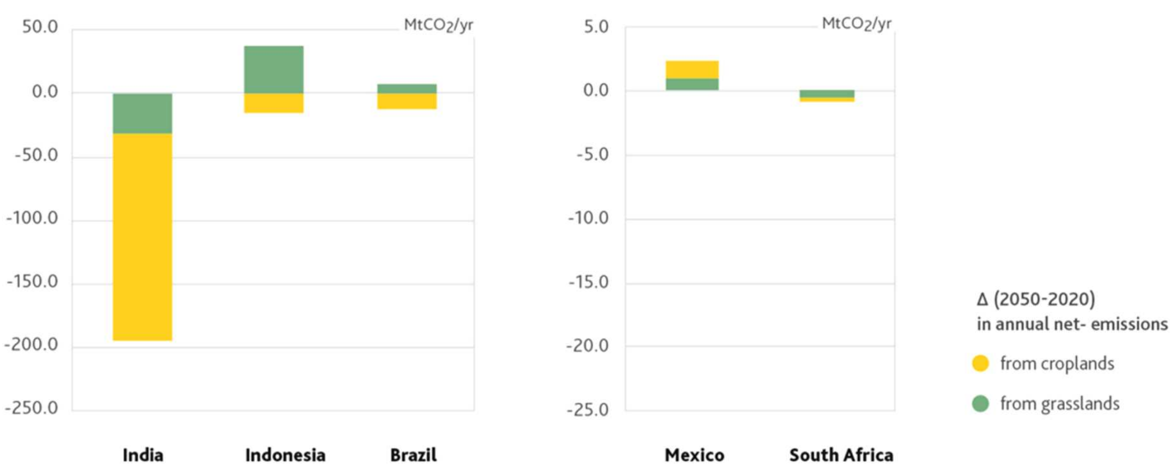

Figure 6. Difference between annual net-emission reductions from agricultural lands in 2050 and 2020, $MtCO_2$.

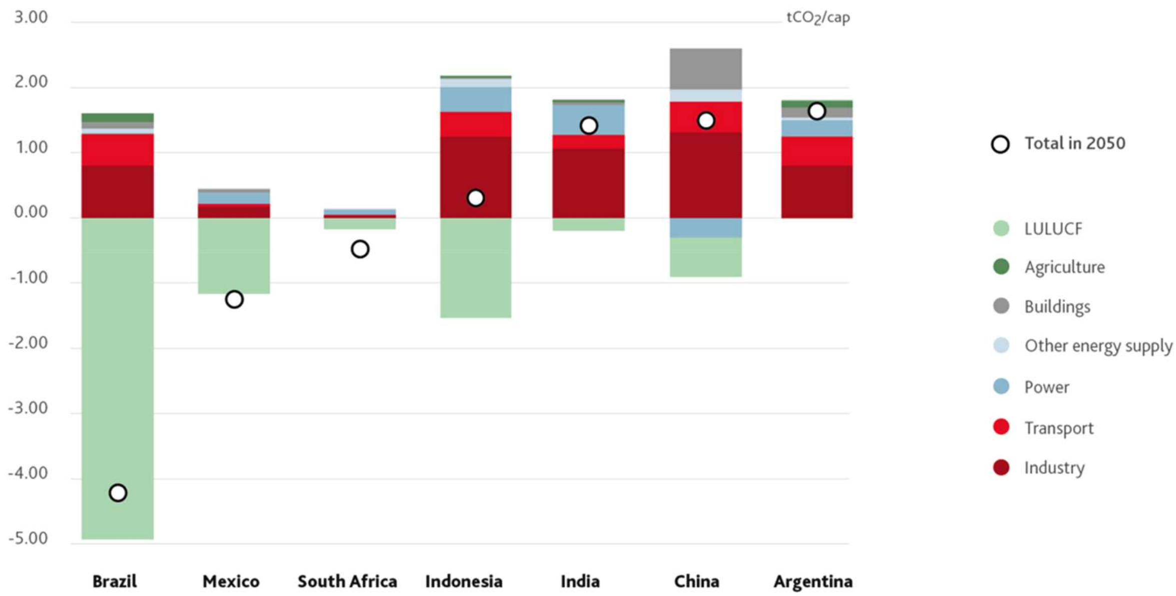

Figure 7. Residual emissions per capita, 2050, tCO_2/cap.

However, by the horizon of national carbon neutrality, most countries reach levels like Mexico and South Africa in 2050, below 0.5 tCO2/cap.

In the industrial sector, their elimination will depend on the speed with which affordable near zero emitting industrial processes are developed. Process-related emissions due to the production of clinker for cement and the underdeveloped nature of substitute binders for Ordinary Portland cement will likely require the use of carbon capture and storage solutions (CCS). Beyond this specific sectoral use and due to the constraints on its potential and challenges affecting the deployment timelines (Annex 4), CCS will play a limited role in mitigation strategies to achieve carbon neutrality. Substitution of novel cementitious materials, for example mixes of ground limestone and calcined clays, can reduce clinker need by up to half for most uses (Bashmakov et al., 2022; Bataille et al., 2023).

In the transport sector, elimination of GHGs will depend on the shift to electric vehicles and biofuels, as well as a shift to less car- and truck-dependent mobility systems for people and goods. In 2050, freight-related emissions represent more than passenger-related emissions. These emissions will critically depend on structural changes in production, consumption and trade patterns that set the constraints for logistics organizations, vehicle technologies and related emissions. These structural changes include: rethinking product and packaging design to lower mass and volumes transported, relocating industrial and storage facilities to shorten distances and facilitate electrification of road vehicles, relaxing logistics constraints on delivery speeds, frequencies and budget to facilitate the consolidation of flows, modal shifts, and optimization of vehicle loads (Briand et al., 2024; Harry-Villain et al., 2025).

In electricity generation, emissions become inconsequential in all countries by their carbon neutrality date, despite a strong increase of demand, because deep decarbonization of power generation is a critical component of any carbon neutrality strategy (Vishwanathan et al., 2025). China even reaches net negative emissions by 2050 due to massive expansion of bioenergy with capture carbon and storage (BECCS).

4. Policy discussion

This section presents and discusses policy recommendations to support the detailed transformations to net-zero as described in Section 3. These policy recommendations are derived from a cross-cutting analysis of the national scenarios as well as from specific national case studies exploring more in-depth some of the policy aspects of the transition.

4.1. Triggering short-term emission reductions in power generation, passenger transport and land use sectors

According to national net-zero pathways, power generation, passenger transport, some industrial actions like cementitious material substitution, and land use sectors have the greatest potential for rapid emission reductions in the next decade. In each of these three sectors technical solutions for CO2 emission reductions are readily available and commercially viable: renewable power generation assets (e.g. photovoltaic, wind, run-of-river hydropower paired with batteries and more transmission), non-motorised and public transport, electric vehicles (EVs) and biofuels, protecting and developing crop-, grass-, and forestland sink capacities. The rapid deployment of these solutions essentially depends on changes in the organizations of sectors, through country-specific policies and actions creating an adequate enabling environment.

In power generation, the accelerated penetration of renewable energy power production can be enabled by adapting market access rules. For example, in some countries like in Mexico, permitting independent private producers to participate in the power market directly or through joint ventures can help attract foreign private capitals, bring new technologies and methods, and favour competition and lower production costs. In other countries like in Argentina, establishing control institutions and clear incentives are key to ensure that private and public power producers align investments and choices with public decisions. This could require creating a public planning agency, to define renewable production quotas, economic subsidies and economic penalties for producers not respecting quotas.

For passenger transport, the accelerated shift to public and non-motorised transport (NMT) can be enabled by specific infrastructure work and transport regulations. For example, in Mexico, reallocating existing road lanes to dedicated bus or NMT lanes, reducing speed limits for cars in cities, reducing public transport ticket price, increasing public transport service punctuality and frequency, can increase the safety, the quality and providing competitive advantages in terms of travel time and cost. The substitution of internal combustion vehicles by electric vehicles can be accelerated by incentivizing local manufacturing through tax exemptions or reduced importation taxes, by defining sales quotas for car manufacturers and procurement renewal quotas for public and large company fleets to establish first niche markets, in setting targets for the distribution system operators to establish public charging stations, or in lowering the upfront cost of EVs for citizens through tax exemptions, like in South Africa and China.

For land use, the accelerated protection of forestlands and limitations of peatland fires would require a real enforcement of existing laws associated with control policies like in Indonesia and Brazil. For example, control agencies can be created and have specific control and enforcement rights. Economic incentives, such as the implementation of a carbon market, an updated system for payments for ecosystem services for land managers or for farmers, would also be key to finance reforestation projects, agroforestry and changes in forest management practices.

For industry, a substantial reduction can be achieved by transforming construction practices. For example, in Mexico or Brazil upgrading building codes and material production processes while training architectural, engineering and construction professionals to use steel and concrete only where needed and most effectively, along with a wider range of other building materials and cementitious material substitutes (e.g. ground limestone & calcined clays), can deliver up to 50% or more emission reductions, compared to current practices. For other industries, substantial emissions reductions are also available with higher and better sorted rates of metal recycling, especially for iron, aluminium and copper.

4.2. Addressing inertias to enable deep emission reductions in the long-term

The analysis of national net-zero pathways shows the importance of preparing adequate conditions for long-term transformations and emission reductions. This means recognizing that decisions delivering short-term emission reductions can also create a risk of lock-in limiting the capacity for longer-term deep emission reduction. This means also considering the actions which may not deliver emissions reductions in the short-term but are essential to achieve the structural and systemic changes necessary to achieve deeper emission reductions by mid-century. Some of these drivers require more time to translate into effective emission reductions because of stronger inertias and actors' resistance to change (Li & Strachan, 2017) like infrastructure and technologies (Fisch-Romito et al., 2021), governance and institutions, lifestyles and behaviours (Seto et al., 2016).

Our results highlight that all sectors are affected by these structural changes and that the sources of inertia require better anticipation and sector and country-specific actions in the short-term. Net-zero pathways require a systemic shift to enable development patterns meeting domestic socio-economic objectives with less energy. This shift goes beyond simply replacing energy-consuming systems with more efficient options, it requires changes in lifestyle and behaviours, such as in the Indian Lifestyle for Environment (LiFE) programme (MoEFCC, 2022b), which include about 30 actionable items in the energy and AFOLU, or actions to support teleworking and car-pooling like in Indonesia, or to subsidies public transport to incentivize modal shift like in Mexico. In the power sector and other energy supply sectors, some countries can be limited in the development and penetration of renewables due to infrastructural inertias that have not been tackled, like the development of power grid and storage infrastructures in Brazil, or the transition of heating infrastructures and systems like in northern China. In these situations, countries may be tempted to build new fossil fuel power plants which would become stranded assets in the long-term, delaying carbon neutrality. In freight transport, reaching net-zero will require the development of non-road logistics, through rail, inland water and coastal transport (Briand et al., 2024). This will depend on large and critical investments in infrastructures like in South Africa or in India, e.g. dedicated freight corridors. But we often forget that as for all sectors with large infrastructure needs, this will require efficient and transparent governance and solid institutions to plan, finance and regulate development and participation rules and rights for all actors over decades, like in South Africa, or changes in existing rules and user incentives to promote intermodality, like in Brazil (Briand et al., 2024).

4.3. Managing the socio-economic components of the transition

Socio-economic challenges driven by price increases can undermine the affordability of basic needs (e.g. food, energy and transport) in the absence of short-term substitution opportunities. Such price changes can occur even without climate policies; however, ambitious climate action, if not managed carefully, can trigger rapid price surges alongside constraints on carbon, particularly affecting energy and food production. These effects disproportionately affect the disadvantaged and vulnerable and can therefore exacerbate inequalities and undermine the acceptability of the low-carbon transition. To address this situation, domestic policy packages must be put in place that combine mitigation instruments with targeted measures designed to address the undesirable social effects. A typical example is the use of the carbon tax revenues to reduce labour taxes or provide direct transfers to the most affected households (Wills et al., 2022). These economic effects can also be managed through enhanced international cooperation. For example, international policies that help derisk financial instruments or develop soft loans could contribute to lowering the cost of investments in the low-carbon transition and facilitate access to low-carbon options at reduced costs (Matthäus & Mehling, 2020).

At the macroeconomic level, the transition to net-zero necessitates structural shifts in national economic systems, which can impact employment, production assets, and trade. This is especially notable with respect to trade imbalances, risks of deindustrialization, public debt, and the need to generate foreign currency to pay for imports, especially for countries heavily reliant on fossil fuel export revenues today. New dependencies could also emerge in accessing low-carbon technologies if local economies are unable to produce them domestically. However, the transition towards carbon neutrality can create jobs and economic opportunities for some countries if they also implement changes in their domestic industrial structure while decarbonizing (Tatham et al., 2024). A diverse range of economic sectors could be involved depending on country circumstances and priorities, such as the mineral value chain industries in Indonesia, solar PV value chains and other renewable energy components in Brazil, Mexico, Indonesia, South Africa, and Argentina, chemical industries in Argentina, or the iron and steel industry in South Africa (Trollip et al., 2022). To implement such industrialization and effectively guide investments, countries can adopt a variety of public policies suited to their specific national circumstances. These may include public procurement with local content requirements, special economic zones, subsidies and investment incentives, public loans, importation bans and duties, regional integration and workforce education programmes.

Altogether, this analysis highlights decarbonization policies as a combination of instruments and actions encompassing financial and fiscal economic policies and other socio-economic measures, to be articulated consistently and tailored to the unique circumstances of each country.

5. Conclusions

This paper presents and analyses national pathways towards net-zero carbon emissions by around mid-century or soon after in seven large emerging economies: Argentina, Brazil, China, India, Indonesia, Mexico and South Africa. The definition and modelling of these national pathways to net-zero has been conducted by in-country experts adopting the pathways design framework of the Deep Decarbonization Pathways initiative, refined to make it relevant to explore the profound, multi-sectoral transformations required by carbon neutrality with sufficient granularity to derive concrete policy insights. Such scientifically robust, nationally anchored, and long-term analysis are critical to facilitate policy discussion on key short-term actions compatible with national long-term climate and development objectives (Dubash, 2023).

The cross-country comparison demonstrated that each country could define its own pathway towards carbon neutrality, considering their national circumstances and in a way that preserves key domestic socio-economic priorities. This analysis highlights that national pathways to net-zero can follow different patterns depending on the starting point of the country in terms of emissions and development and on the target date for carbon neutrality. But, in all cases, the transition requires profound transformations of energy supply and demand, which notably involves a continuous decrease in the use of fossil fuels and reductions of the energy intensity of economic activity as compared to current policies. This should be established in a country-driven manner to make them compatible with ensuring energy access and delivering basic energy services. All net-zero pathways also entail pro-active action on the land use sector to deliver net negative emissions while preserving other socio-economic dimensions (Svensson et al., 2024).

To trigger these net-zero transformations, short-term policy packages should be adjusted to each country context and combine three main components. First, they should create an enabling environment and adapt the current sectors' organization for the fast deployment of readily available and economically profitable solutions in power generation, passenger transport, and land use, where most of the short-term emission reduction can be reached. Second, they should prepare longer-term emission reductions by addressing critical inertias related to changes in infrastructures, technologies, lifestyles and institutions. Finally, they should address the socio-economic and macroeconomic aspects of the transition by managing undesirable effects on inequalities and accompanying structural changes in the economic systems.

This work enabled us to identify some limitations and areas for improvement in future national pathways. First, the consideration of demand-side systemic mitigation options for industrial and freight transport and a better articulation of land use with energy system changes would require further refinements. Second, the analysis of non-CO2 emissions from fugitive sources in the energy system or from the agriculture sector has not always been well reported, due to a lack of high-quality data. Finally, socio-economic analysis is critical, both to assess macroeconomic or micro-economic indicators. In the last case, this project initiated first research collaborations to improve modelling capacities in that field but developing a more multidisciplinary work between economic and finance modelling experts and climate experts would also be key in the future, as suggested by other initiatives, like the Green Macroeconomic Modeling Initiative (GMMI), the Coalition for Capacity on Climate Action (C3A) or the Task Force for the Global Mobilization against Climate Change launched by the G20 Brazilian presidency.

Finally, the national pathways developed in this project could also serve to produce a cross-country comparison of considered measures and related barriers or enablers, of key inertias in the different sub-sectors, and of the role of international cooperation.

Acknowledgements

The authors thank and acknowledge Ivan Pharabod for design of our figures.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

IDDRI gratefully acknowledges its ongoing financial support from the French National Research Agency, Grant Number: ANR-10-LABX-14-01, the Michelin Corporate Foundation and the International Climate Initiative (IKI) of the German Federal Ministry for the Environment, Nature Conservation and Nuclear Safety (BMU) through the '2050 is now: Aligning climate action with long-term climate and development goals' project.

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References

Barrett, J., Pye, S., Betts-Davies, S., Broad, O., Price, J., Eyre, N., Anable, J., Brand, C., Bennett, G., Carr-Whitworth, R., Garvey, A., Giesekam, J., Marsden, G., Norman, J., Oreszczyn, T., Ruyssevelt, P., & Scott, K. (2022). Energy demand reduction options for meeting national zero-emission targets in the United Kingdom. *Nature Energy*, *7*(8), 726–735. https://doi.org/10.1038/s41560-022-01057-y

Bashmakov, I., Nilsson, L., Acquaye, A., Bataille, C., Cullen, M., de la Rue du Can, S., Fischedick, M., Geng, Y., Tanaka, K., Bauer, F., Hasanbeigi, A., Levi, P., Myshak, A., Perczyk, D., Philibert, C., & & Samadi, S. (2022). IPCC AR6 WGIII Chapter 11 Industry. In Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. https://doi.org/10.5743/cairo/9789774165603.003.0014.

Bataille, C., Stiebert, S., Hebeda, O., Trollip, H., McCall, B., & Vishwanathan, S. S. (2023). Towards net-zero emissions concrete and steel in India, Brazil and South Africa. *Climate Policy*, https://doi.org/10.1080/14693062.2023.2187750

Bataille, C., Waisman, H., Briand, Y., Svensson, J., Vogt-Schilb, A., Jaramillo, M., Delgado, R., Arguello, R., Clarke, L., Wild, T., Lallana, F., Bravo, G., Nadal, G., Le Treut, G., Godinez, G., Quiros-Tortos, J., Pereira, E., Howells, M., Buira, D., … Imperio, M. (2020). Net-zero deep decarbonization pathways in Latin America: Challenges and opportunities. *Energy Strategy Reviews*, *30*. https://doi.org/10.1016/j.esr.2020.100510

Bataille, C., Waisman, H., Colombier, M., Segafredo, L., Williams, J., & Jotzo, F. (2016). The need for national deep decarbonization pathways for effective climate policy. *Climate Policy*, *16*(June), S7–S26. https://doi.org/10.1080/14693062.2016.1173005

Braunreiter, L., van Beek, L., Hajer, M., & van Vuuren, D. (2021). Transformative pathways – using integrated assessment models more effectively to open up plausible and desirable low-carbon futures. *Energy Research and Social Science*, *80*(January), 102220. https://doi.org/10.1016/j.erss.2021.102220

Briand, Y., Gupta, D., D'Agosto, M. d. A., Goes, G. V., Gonçalves, D. N. S., Garg, A., Vishwanathan, S. S., Ahjum, F., Trollip, H., McCall, B., Siagan, U. W. R., Dewi, R. G., Pye, S., Combes, F., & Koning, M. (2024). A pathway design framework for national freight transport decarbonization strategies. *Climate Policy*, https://doi.org/10.1080/14693062.2024.2412709

Briand, Y., Pye, S., D'Agosto, M. d. A., Goes, G. V., Neves Schmitz-Gonçalves, D., Garg, A., Gupta, D., Vishwanathan, S. S., Siagian, U. W. R., Ahjum, F., & Trollip, H. (2023). Passenger transport decarbonization in emerging economies: Policy lessons from modelling long-term deep decarbonization pathways. *Climate Policy*, 1–21. https://doi.org/10.1080/14693062.2023.2194859

Català, A., Waisman, H., Torres Gunfaus, M., Jaramillo, M., & Achaichia, S. (2024). NDC-LT-LEDS alignment guide aligning short-term plans with long-term ambitions guide. https://www.iddri.org/en/publications-and-events/report/ndc-lt-leds-alignment-guide-aligning-short-term-plans-long-term.

Creutzig, F., Niamir, L., Bai, X., Callaghan, M., Cullen, J., Díaz-José, J., Figueroa, M., Grubler, A., Lamb, W. F., Leip, A., Masanet, E., Mata, É, Mattauch, L., Minx, J. C., Mirasgedis, S., Mulugetta, Y., Nugroho, S. B., Pathak, M., Perkins, P., … Ürge-Vorsatz, D. (2021). Demand-side solutions to climate change mitigation consistent with high levels of well-being. *Nature Climate Change*, *12*(1), 36–46. https://doi.org/10.1038/s41558-021-01219-y

Creutzig, F., Roy, J., Lamb, W. F., Azevedo, I. M. L., Bruine De Bruin, W., Dalkmann, H., Edelenbosch, O. Y., Geels, F. W., Grubler, A., Hepburn, C., Hertwich, E. G., Khosla, R., Mattauch, L., Minx, J. C., Ramakrishnan, A., Rao, N. D., Steinberger, J. K., Tavoni, M., Ürge-Vorsatz, D., & Weber, E. U. (2018). Towards demand-side solutions for mitigating climate change. *Nature Climate Change*, *8*(4), 268–271. https://doi.org/10.1038/s41558-018-0121-1

DDPP. (2015). Pathways to deep decarbonization 2015 report. IDDRI & SDSN.

de Bortoli, A., Chanel, A., Chabas, C., Greffe, T., & Louineau, E. (2025). More rationality and inclusivity are imperative in reference transition scenarios based on IAMs and shared socio-economic pathways - recommendations for prospective LCA. *Renewable and Sustainable Energy Reviews*, *222*(May), 115924. https://doi.org/10.1016/j.rser.2025.115924

Dubash, N. K., Mitchell, C., Boasson, E. L., Borbor-Cordova, M. J., Fifita, S., Haites, E., Jaccard, M., Jotzo, F., Naidoo, S., Romero-Lankao, P., Shlapak, M., Shen, W., & Wu, L. (2022). National and sub-national policies and institutions. In *IPCC, 2022: Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press. https://doi.org/10.1017/9781009157926.015.
Dubash, N. K. (2023). Rebalance attention from global target setting toward national climate politics and policy. *Science*, *382*, 6666. https://doi.org/10.1126/SCIENCE.ADK7428
Fisch-Romito, V., Guivarch, C., Creutzig, F., Minx, J. C., & Callaghan, M. W. (2021). Systematic map of the literature on carbon lock-in induced by long-lived capital. *Environmental Research Letters*, *16*(5), https://doi.org/10.1088/1748-9326/aba660
Garg, A., Patange, O., Vishwanathan, S. S., Nag, T., Singh, U., & V, A. (2024). Synchronizing energy transitions toward possible Net-Zero for India: Affordable and clean energy for all Indian Institute of Management Ahmedabad.
Government of Brazil. (2024). Second Nationally Determined Contribution of Brazil. https://unfccc.int/sites/default/files/2024-11/Brazil_Second Nationally Determined Contribution %28NDC%29_November2024.pdf.
Government of China. (2021). China's mid-century long-term low greenhouse gas emission development strategy. https://unfccc.int/sites/default/files/resource/China%E2%80%99s%20Mid-Century%20Long-Term%20Low%20Greenhouse%20Gas%20Emission%20Development%20Strategy.pdf.
Government of Indonesia. (2021). Indonesia long-term strategy for low carbon and climate resilience 2050. In Minister of Environment and Forestry.
Government of South Africa. (2020). South Africa's low emission development strategy 2050. https://unfccc.int/sites/default/files/resource/South%20Africa%27s%20Low%20Emission%20Development%20Strategy.pdf.
Harry-Villain, L., Briand, Y., Waisman, H., Rudolph, F., Hall, C., van Asselt, H., Gjerek, M., Merkert, R., de Almeida D'Agosto, M., Goes, G. V., Neves Schmitz Gonçalves, D., Cadena, R. D., Munoz, M. R., Gupta, D., Hassani, A., Tovilla, J., Hernandez, T., Emodi, N. V., Jensen, S. A., … Cui, R. (2025). Analyzing sectoral policies for deep decarbonization: The case of national freight transportation. *Climate Policy*, 1–14. https://doi.org/10.1080/14693062.2025.2485192
IPCC. (2022). IPCC AR6 WG III - summary for policymakers. In Climate Change 2022 - mitigation of climate change. https://doi.org/10.1017/9781009157926.
Lefèvre, J., Briand, Y., Pye, S., Tovilla, J., Li, F., Waisman, H., Cayla, J., Zhang, R., Lefèvre, J., Briand, Y., Pye, S., Tovilla, J., & Li, F. (2020). A pathway design framework for sectoral deep decarbonization : The case of passenger transportation. *Climate Policy*, *0*(0), 1–14. https://doi.org/10.1080/14693062.2020.1804817
Li, F. G. N., & Strachan, N. (2017). Modelling energy transitions for climate targets under landscape and actor inertia. *Environmental Innovation and Societal Transitions*, *24*, 106–129. https://doi.org/10.1016/j.eist.2016.08.002
Matthäus, D., & Mehling, M. (2020). De-risking renewable energy investments in developing countries: A multilateral guarantee mechanism. *Joule*, *4*(12), 2627–2645. https://doi.org/10.1016/j.joule.2020.10.011
MoEFCC. (2022a). *India's long-term low-carbon development strategy: Submission to the United Nations Framework Convention on Climate Change*, 1–121.
MoEFCC. (2022b). *Mission lifestyle for environment (LiFE)*. Government of India. https://missionlife-moefcc.nic.in/.
Muttitt, G., Price, J., Pye, S., & Welsby, D. (2023). Socio-political feasibility of coal power phase-out and its role in mitigation pathways. *Nature Climate Change*, *13*(2), 140–147. https://doi.org/10.1038/s41558-022-01576-2
Nabuurs, G. J. (2022). Agriculture, forestry and other land uses (AFOLU). In Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change (pp. 747–860). Cambridge University Press. https://doi.org/10.1017/9781009157926.009.
Pye, S., & Bataille, C. (2016). Improving deep decarbonization modelling capacity for developed and developing country contexts. *Climate Policy*, *16*, S27–S46. https://doi.org/10.1080/14693062.2016.1173004
Robinson, J. B. (1982). Energy backcasting a proposed method of policy analysis. *Energy Policy*, *10*(4), 337–344. https://doi.org/10.1016/0301-4215(82)90048-9
Seto, K. C., Davis, S. J., Mitchell, R. B., Stokes, E. C., Unruh, G., & Ürge-Vorsatz, D. (2016). Carbon lock-in: Types, causes, and policy implications. *Annual Review of Environment and Resources*, *41*, 425–452. https://doi.org/10.1146/annurev-environ-110615-085934
Svensson, J., Avashia, V., Boer, R., Chaturvedi, R. K., Cotta, M., Dubeux, C., Sugih Immanuel, G., La Rovere, E. L., Patange, O., Rossita, A., & Vishwanathan, S. S. (2024). The AFOLU sector's role in national decarbonization: A comparative analysis of low-GHG development pathways in Brazil. *India and Indonesia. Climate Policy*, 1–18. https://doi.org/10.1080/14693062.2024.2391048
Svensson, J., Waisman, H., Vogt-Schilb, A., Bataille, C., Aubert, P. M., Jaramilo-Gil, M., Angulo-Paniagua, J., Arguello, R., Bravo, G., Buira, D., Collado, M., De La Torre Ugarte, D., Delgado, R., Lallana, F., Quiros-Tortos, J., Soria, R., Tovilla, J., & Villamar, D. (2021). A low GHG development pathway design framework for agriculture, forestry and land use. *Energy Strategy Reviews*, *37*(September 2020), 100683. https://doi.org/10.1016/j.esr.2021.100683
Tatham, J., Burton, J., Merven, B., Mcdaid, N., Mccall, B., Cunliffe, G., & Caetano, T. (2024). Assessing the economy-wide effect of localising renewable energy value chains in South Africa. *March*, 1–21.
Trollip, H., McCall, B., & Bataille, C. (2022). How green primary iron production in South Africa could help global decarbonization. *Climate Policy*, *22*(2), 236–247. https://doi.org/10.1080/14693062.2021.2024123
UNEP. (2024). Emissions Gap Report 2024: No more hot air … please! .
UNFCCC. (2015). Paris Agreement. In Paris Agreement. https://unfccc.int/sites/default/files/english_paris_agreement.pdf.
UNFCCC. (2021). Glasgow climate pact. https://unfccc.int/documents/310475.
UNFCCC. (2023). Outcome of the first Global Stocktake. https://unfccc.int/sites/default/files/resource/cma2023_16a01E.pdf.

Vishwanathan, S. S., Bataille, C., Nogueira, E. C., McCall, B., Siagian, Ucok W. R., Cunha, S., Lebre La Rovere, E.,Trollip, H., Garg, A., Patange, O., Burton, J., & Torres Gunfaus, M. (2025). Towards net-zero electricity sector in emerging economies – Brazil, India, Indonesia and South Africa.

Waisman, H., Bataille, C., Winkler, H., Jotzo, F., Shukla, P., Colombier, M., Buira, D., Criqui, P., Fischedick, M., Kainuma, M., La Rovere, E., Pye, S., Safonov, G., Siagian, U., Teng, F., Virdis, M. R., Williams, J., Young, S., Anandarajah, G., … Trollip, H. (2019). A pathway design framework for national low greenhouse gas emission development strategies. *Nature Climate Change*, *9*(4), 261–268. https://doi.org/10.1038/s41558-019-0442-8

Wiese, F., Taillard, N., Balembois, E., Best, B., Bourgeois, S., Campos, J., Cordroch, L., Djelali, M., Gabert, A., Jacob, A., Johnson, E., Meyer, S., Munkácsy, B., Pagliano, L., Quoilin, S., Roscetti, A., Thema, J., Thiran, P., Toledano, A., … Marignac, Y. (2024). The key role of sufficiency for low demand-based carbon neutrality and energy security across Europe. *Nature Communications*, *15*(1), 9043. https://doi.org/10.1038/s41467-024-53393-0

Wills, W., La Rovere, E. L., Grottera, C., Naspolini, G. F., Le Treut, G., Ghersi, F., Lefèvre, J., & Dubeux, C. B. S. (2022). Economic and social effectiveness of carbon pricing schemes to meet Brazilian NDC targets. *Climate Policy*, *22*(1), 48–63. https://doi.org/10.1080/14693062.2021.1981212